

Customer Shopping Behaviour Analysis

1. Executive Summary

This report outlines the end-to-end data analysis process completed for Customer Shopping Behaviour Analysis project. The primary objective was to analyse customer behaviour data and deliver actionable business insights. The workflow utilized a modern data analytics stack, leveraging Python for robust data cleaning, SQL for structured data exploration, and Power BI for dynamic, interactive visualizations.

2. Data Summary

Rows: 3,900

Columns: 18

Key Features:

Customer demographics (Age, Gender, Location, Subscription Status)

Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

Missing Data: 37 values in Review Rating column

2. Project Architecture and Workflow

The project was executed in three distinct phases to ensure data integrity and deep analytical depth:

Phase 1: Data Preprocessing and Cleaning using Python.

Phase 2: Structural Data Exploration and In-depth Analysis using SQL.

Phase 3: Interactive Dashboard Development and Reporting via Power BI.

3. Python Preprocessing Pipeline

The raw customer data was transformed into a structured, analysis-ready format through the following steps:

Data Profiling: Loaded the data into a Data Frame, using `read_csv()`, `df.head()`, and `df.info()` to inspect schemas, data types, and null distribution.

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID           3900 non-null   int64
1   Age                   3900 non-null   int64
2   Gender                3900 non-null   str
3   Item Purchased        3900 non-null   str
4   Category              3900 non-null   str
5   Purchase Amount (USD) 3900 non-null   int64
6   Location              3900 non-null   str
7   Size                  3900 non-null   str
8   Color                 3900 non-null   str
9   Season                3900 non-null   str
10  Review Rating          3863 non-null   float64
11  Subscription Status    3900 non-null   str
12  Shipping Type          3900 non-null   str
13  Discount Applied       3900 non-null   str
14  Promo Code Used        3900 non-null   str
15  Previous Purchases     3900 non-null   int64
16  Payment Method         3900 non-null   str
17  Frequency of Purchases 3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB
```

Imputation: Filled missing values in the Review Rating column with the dataset's median rating to maintain statistical integrity without sacrificing rows.

```
df.isnull().sum()
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size             0
Color            0
Season           0
Review Rating     0
Subscription Status  0
Shipping Type     0
Discount Applied  0
Promo Code Used   0
Previous Purchases  0
Payment Method    0
Frequency of Purchases  0
dtype: int64
```

Schema Standardization: Converted all column names to lowercase strings separated by underscores (snake_case) to ensure SQL compatibility and query uniformity.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns = {'purchase_amount_(usd)' : 'purchase_amount'})
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

Data Transformation & Enrichment:

Created an age_group column using pd.qcut() and custom labels to segment customers into equal-sized demographic bins.

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

```
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

Transformed the purchase frequency column from text-based day descriptions into purely numeric values to allow mathematical aggregation.

```
df['frequency_of_purchases'].unique()

<ArrowStringArray>
[ 'Fortnightly',      'Weekly',      'Annually',      'Quarterly',
  'Bi-Weekly',      'Monthly', 'Every 3 Months']
Length: 7, dtype: str

frequency_mapping = {
    'Fortnightly' : 14, 'Weekly' : 7, 'Annually' : 365, 'Quarterly' : 90, '
    'Bi-Weekly' : 14, 'Monthly' : 30, 'Every 3 Months' : 90}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

Deduplication: Dropped the promo_code_used column because it was completely redundant with the discount_applied column, optimizing storage and data clarity.

```
(df['discount_applied'] == df['promo_code_used']).all()

np.True_

df= df.drop('promo_code_used', axis=1)

df.columns

Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='str')
```

4. Database Ingestion

The finalized, cleaned Data Frame was exported and loaded into the SQL customer database, establishing a structured table optimized for complex queries and Power BI integration.

5. Phase 2: Exploratory Data Analysis & Business Intelligence (SQL)

Using the clean database table, structured SQL queries were executed to extract business metrics regarding revenue performance, customer segmentation, and product trends. The exploration solved the following ten business questions:

Q1. Demographic Revenue Split (SUM, GROUP BY): Calculated the total revenue generated by male vs. female customers to determine core buying segments.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Q2. High-Value Discount Shoppers (Subqueries): Isolated individual customers who utilized a discount code but still spent more than the overall average purchase amount.

Q3. Top-Rated Product Lines (AVG, LIMIT): Identified and ranked the top 5 specific items based on their average review ratings to gauge customer satisfaction.

	item_purchased text	average_review_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Q4. Logistics Cost Comparison (Filtering): Compared average transaction amounts specifically between "Standard" and "Express" shipping methods to evaluate premium service adoption.

	shipping_type text	average_purchase_amount numeric
1	Standard	58.46
2	Express	60.48

Q5. Subscription Financial Value (Multi-Metric Aggregation): Compared the total customer counts, average spend, and lifetime revenue between active subscribers and non-subscribers.

	subscription_status text	total_customer bigint	average_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645
2	No	2847	59.87	170436

Q6. Product Discount Rates (CASE WHEN Ratio): Calculated the exact percentage of orders that used a discount for each product, pinpointing the top 5 most heavily discounted items.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Q7. Customer Loyalty Segmentation (Common Table Expressions - CTE): Grouped shoppers into New (1 purchase), Returning (2–10 purchases), and Loyal (>10 purchases) cohorts based on their historical transaction counts.

	customer_segment text	Number of customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Q8. Top 3 Products Per Category (ROW_NUMBER() Function): Partitioned the data by retail category to rank and extract the top 3 most-purchased items inside each individual category bucket.

Q9. Behavioral Subscription Correlation (Behavioral Filtering): Analysed repeat buyers (customers with more than 5 previous purchases) to see if high transaction frequency correlates with higher subscription rates.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

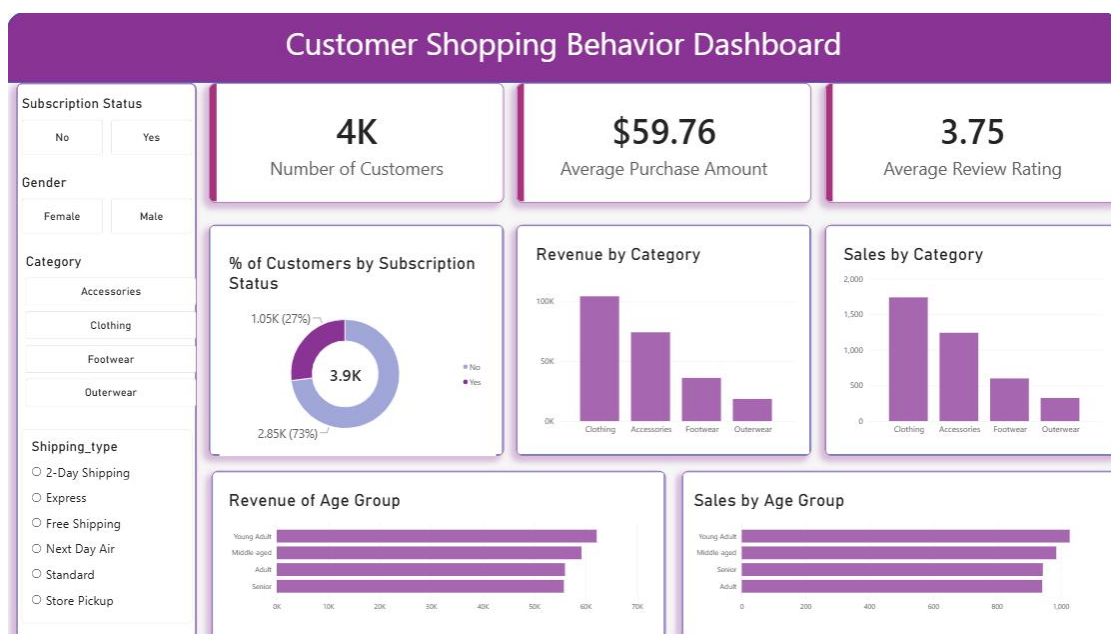
Q10. Age Group Revenue Contribution (Engineered Grouping): Aggregated total sales volumes against the custom age_group brackets created during the Python preprocessing stage.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

6. Phase 3: Data Visualization & Reporting (Power BI)

A clean, interactive "Customer Shopping Behaviour Dashboard" was built to turn this raw data into clear visual stories across five key areas:

- **Key Performance Indicators (KPIs):** Displays the main business health stats at a single glance, showing 4,000 total customers, an average spend of \$59.76, and a 3.75 out of 5 average review rating.
- **Subscription Breakdown:** Uses a clear donut chart to show loyalty splits, proving that the vast majority of shoppers are non-subscribers (73% or 2.85K) compared to active subscribers (27% or 1.05K).
- **Category Performance:** Uses vertical bar charts to easily compare different product lines, revealing that Clothing brings in the most sales and highest revenue out of all categories.
- **Age Demographics:** Uses horizontal bar charts to track spending against our custom age groups, clearly pinpointing "Young Adults" as our most profitable customer group.
- **Interactive Filters (Slicers):** Includes an easy-to-use left sidebar menu where anyone can instantly filter all the charts by *Subscription Status*, *Gender*, *Category*, or *Shipping Type* with one click.



7. Data-Driven Business Recommendations

Target the "Young Adult" Segment: Focus marketing budgets on this group as they generate your highest overall revenue.

Convert Non-Subscribers: Launch sign-up perks targeting the 73% non-subscriber majority to quickly grow recurring revenue.

Bundle with Clothing: Package lower-performing product categories (Accessories/Footwear) alongside top-selling Clothing items to increase inventory movement.

Increase Average Order Value (AOV): Set a free shipping threshold at \$65 or \$70 to organically push the current \$59.76 average spend baseline higher.